PARKSHARK: SMART CAMPUS MICRO-MOBILITY & IOT BICYCLE MANAGEMENT SYSTEM

Comprehensive Project Demonstration & Executive Technical Report | VIT Vellore

Project: ParkShark IoT Smart Rack Platform	**Institution:** VIT Vellore (372 Acres)
Author / Lead: Tanishka Sahu (21BCE1001)	**Tech Stack:** React 19, Three.js, Cloud Functions, Firestore
GitHub: github.com/TanishkaSahu191/ParkShark	**Automated Tests:** 105 passing (100% pass rate)

1. Problem Statement & Campus Motivation

With over 35,000 students across 372 acres, bicycle commute is essential at VIT Vellore. However, during morning peak class hours (8:00–9:00 AM), intense rack shortages lead to chaotic parking that blocks fire escape doors, wheelchair ramps, and walkways. Mechanical locks suffer rampant clipping and zero theft accountability. Students waste 10–15 minutes hunting for racks, while security administrators lack data on slot utilization and maintenance.

2. The ParkShark Solution Architecture

ParkShark unifies physical IoT racks with an interactive 3D digital twin of the VIT campus. Key capabilities include:

- **3D Digital Twin Engine:** Accurate 1:1 scale (1 unit = 1.0 m) mapping of Academic Blocks AB1, AB2, and AB3 with 150 smart racks and live availability status.
- **Waypoint Ground Navigation:** Dynamic route pathfinding with real-time distance telemetry and human walking pace estimates.
- **Cryptographic QR Locking:** Signed QR code matrix verification with HMAC-SHA256 salted PIN encryption and 5-minute brute-force lockout safeguards.
- **Admin Command Center:** Campus-wide occupancy metrics, cold-to-hot 3D heatmap analytics, remote force-release, and printable QR batch utilities.
- **Grievance Portal:** In-app ticketing system with photo upload and real-time status resolution tracking.

3. Technical Specifications & Stack

Layer	Technologies Used	Engineering Function
Frontend & PWA	React 19, TypeScript, Vite, Tailwind v4	Progressive Web App with custom Service Worker and offline fallback
3D Digital Twin	Three.js, React Three Fiber, Drei	60 FPS campus walkthrough, 2D radar minimap, dynamic compass
Backend / Cloud	Google Cloud Functions v2, App Check	Serverless callable endpoints with sliding-window rate limiting
Database	Cloud Firestore, PGLite WebAssembly	Atomic transaction updates, student data isolation rules, local cache
Security	Node.js Crypto, HMAC-SHA256, reCAPTCHA	Salted PIN hashing, brute force lockout, Phone OTP auth
Testing	Node Test Runner, Playwright E2E	105 automated unit/integration tests with zero failures

4. Real-World Impact & Live Demonstration Access

- **Walkway Safety:** Completely eliminates haphazard parking and clears campus emergency corridors.
- **Zero Theft:** Cryptographic PIN logs replace untraceable mechanical padlocks.
- **Sustainability:** Removes commuting friction to boost zero-emission cycling over shuttle buses.
- **GitHub Repository:** github.com/TanishkaSahu191/ParkShark
- **Local Live Server:** http://localhost:8080/ (Fully functional & tested)
- **Demo Credentials:** Student ID: **21BCE1001** | Admin ID: **ADMIN001** (Test OTP: 123456)